

Question #: 1

A 40-year-old man comes to the physician because of a 2-week history of nausea, vomiting, and abdominal pain. Physical examination shows no abnormalities. A gastric biopsy obtained from endoscopy shows a patchy erosion of the epithelial lining with mucosal infiltration by lymphocytes, macrophages, and plasma cells. A diagnosis of chronic peptic ulcer is made. Which of the following best describes this type of inflammation?

- A. Acute suppurative inflammation
- B. Chronic granulomatous formation
- C. Acute fibrinous inflammation
- D. Acute serous inflammation
- ✓E. Chronic non-specific inflammation

Rationale: The chronic inflammatory infiltrate is characterized by the large numbers of lymphocytes, macrophages, and plasma cells in contrast to acute inflammation where the major cell type is the neutrophil. Peptic ulcer disease is a great example of non-specific chronic inflammation and usually arises from non-resolution of acute inflammation.

Chronic granulomatous inflammation is characterized by the clusters of epithelioid cells and multinucleate giant cells. A common example is tuberculosis

Question #: 2

A 77-year-old man is brought to the emergency department from a nursing home because of a 1-day history of decreased consciousness. He was diagnosed with a perineal infection and started treatment with antibiotics 3 days ago. His blood pressure is 90/70 mmHg, pulse is 98/min and temperature is 104°F (40°C). Physical examination shows a febrile, comatose man with a soft mass in the perineal area. Laboratories studies show markedly elevated neutrophils. The physician suspects septicemia secondary to a perineal abscess. Which of the following mediators is most likely responsible for the fever observed in this patient?

- A. Histamine
- ✓B. Prostaglandins
- C. Nitric oxide
- D. Serotonin
- E. Chemokines

Rationale: Prostaglandins and cytokines are mainly responsible for the fever observed in acute inflammatory states. Histamine, nitric oxide, and serotonin play a role in vasodilation and chemokines promote leukocyte activation.

Question #: 3

A 40-year-old man comes to the physician because of a 2-week history of nausea, fever, and chest pain. Physical examination shows an ill-looking, febrile man. Imaging studies show fluid in the pericardial cavity. Analysis of the fluid shows that it is rich in fibrinogen. Which of the following best describes this type of inflammation?

- A. Acute suppurative inflammation
- ✓B. Acute fibrinous inflammation
- C. Acute serous inflammation
- D. Chronic non-specific inflammation
- E. Chronic granulomatous formation

Rationale: The chronic inflammatory infiltrate is characterized by the large numbers of lymphocytes, macrophages, and plasma cells in contrast to acute inflammation where the major cell type is the neutrophil. Chronic granulomatous inflammation is characterized by the clusters of epithelioid macrophages and multinucleate giant cells. The presence of fibrinogen informs that this is a fibrinous inflammation.

Question #: 4

A 5-year-old boy is brought to the emergency department because of a 4-day history of dry cough and shortness of breath. He is a known asthmatic and has been hospitalized for exacerbation of this condition twice in the past 6 months. His blood pressure is 100/70 mm Hg, respirations are 26/min, and temperature is 36°C (96.8°F). Physical examination shows the use of accessory muscles and an expiratory wheeze. A diagnosis of acute exacerbation of bronchial asthma is made. Which of the following morphological findings best characterizes this condition?

- A. Increased destruction of elastic fibers
- B. Thickening of the interalveolar septum
- ✓C. Constriction of the smooth muscles of the bronchioles
- D. Squamous metaplasia of respiratory epithelia
- E. Deficiency of surfactant production

Rationale: Asthma is a chronic inflammatory disease involving the airways of the lungs. It is characterized by recurrent obstruction of airflow that results from inflammation of the bronchioles and constriction of their smooth muscles (bronchospasm). Obstruction of airways makes it difficult for air to move in and out of the lung alveoli, causing symptoms such as wheezing, coughing, shortness of breath, and chest tightness. Inflammation of the respiratory mucosa, underlying connective tissue, and smooth muscles of bronchioles is found in patients with asthma. It is characterized by infiltration of the bronchiolar wall by eosinophils (in some cases, neutrophils), lymphocytes (mostly activated helper T cells), and mast cells. In individuals with asthma, the bronchiolar epithelium is thick, contains an increased number of goblet cells (thus producing more mucus), and has a thick basement membrane due to an increased deposition of

collagen fibers in the reticular lamina. The smooth muscle layer is also more pronounced and contains several layers of hyperplastic smooth muscle cells.

Question #: 5

A 35-year-old woman comes to the physician because of a 1-week history of a tender swelling in her right armpit. Physical examination shows enlarged right axillary lymph nodes. A biopsy is obtained from one of the enlarged organs for histopathological examination. Which of the following stains would most appropriately demonstrate the presence of the fibers in the stroma of this organ?

- A. Hematoxylin & Eosin
- B. Trichrome
- C. Sudan black
- D. Giemsa
- ✓E. Silver nitrate
- F. Oil red O

Rationale: The stroma of lymph nodes consists of reticular tissue made up of reticular cells and reticular fibers which can be stained with a silver preparation.

Question #: 6

A 50-year-old man comes to the physician because of a 2-month history of poor appetite. He has a history of a 20-pack year of cigarette smoking. Physical examination shows scattered wheezes in both lung fields. A bronchial biopsy shows stratified squamous non-keratinized epithelium. Which of the following functions will most likely be affected as a result of this epithelial change?

- A. Inability to produce surfactant
- B. Inability to adjust airway diameter
- C. Inability to promote gaseous exchange
- ✓D. Inability to move mucus
- E. Inability to secrete CC16

Rationale: The cilia on the columnar cells of the respiratory epithelium forms the muco-ciliary escalator which moves mucus out of the lungs. Once these cells are replaced in the presence of noxious stimuli, then this function is lost, and the patient compensates by cough. In this case, specifically it is the respiratory epithelium that has affected, because the biopsy was done on the bronchi.

- Option A - Alveolar wall
- Option B - Bronchiole
- Option C - Alveolar wall
- Option E - Bronchiole

Question #: 7

A 35-year-old man was brought to the emergency department because of sudden onset of severe abdominal pain following a car crash 20 minutes ago. Physical examination shows a distended abdomen. An ultrasound of the abdomen shows a ruptured spleen. Which of the following is most likely a characteristic feature of the red pulp of this organ?

- A. Afferent lymphatics
- B. Efferent lymphatics
- C. Continuous capillaries
- D. Fenestrated capillaries
- ✓E. Sinusoidal capillaries

Rationale: The red pulp of the spleen consists of splenic cords and splenic sinuses. The sinuses are special sinusoidal or discontinuous capillaries.

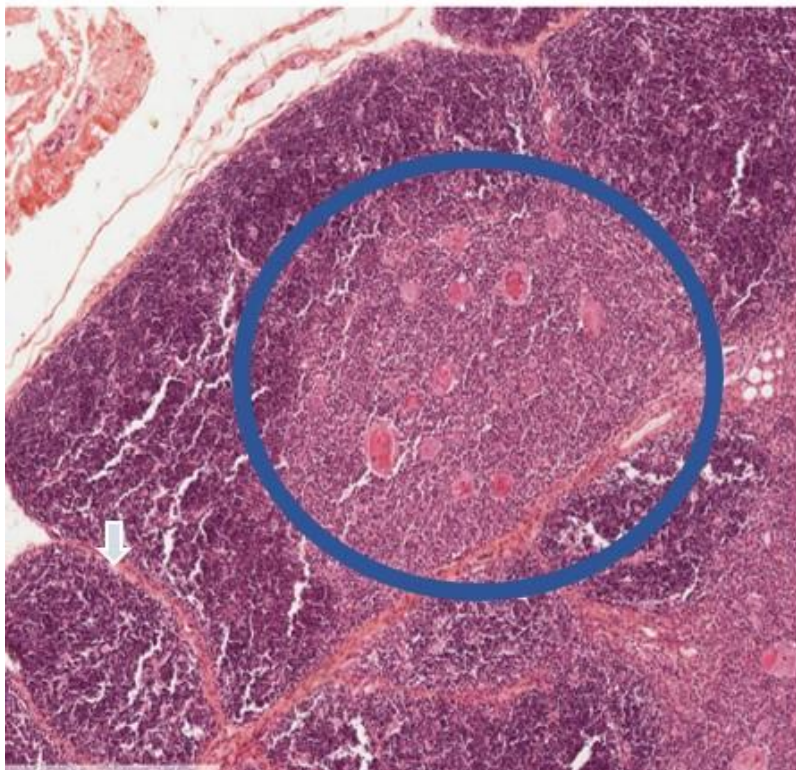
Question #: 8

A 3-year-old boy is brought to the physician by his father because of a 2-hour history of shortness of breath. He has a history of recurrent infections of the respiratory tract. He is suspected of suffering from a deficiency of protective lymphocytes which exposes him to both intracellular and extracellular infections. A stained section of the affected organ is shown in the attached photomicrograph. Which of the following best represents the cell that will most likely give rise to the eosinophilic corpuscles in the highlighted region of the affected organ?

- A. B-lymphocytes
- B. T-lymphocytes
- C. Reticular cells
- ✓D. Epithelioreticular cells
- E. Plasma cells

Rationale: Epithelioreticular cells provide a framework for developing T cells in the thymus. In the medulla they become closely packed forming the concentric thymic corpuscles.

Attachment:



thymus .jpg

Question #: 9

A 49-year-old man comes to the physician because of a 2-month history of loss of appetite and unintentional weight loss. He has a 20-pack-year history of cigarette smoking. Physical examination shows a productive cough with sputum. A biopsy specimen of bronchial mucosa shows stratified squamous non-keratinized epithelium. Which of the following best describes the normal epithelium of this part of the airway?

- A. Stratified columnar
- B. Transitional
- C. Simple squamous
- ✓D. Pseudostratified
- E. Simple cuboidal

Rationale: The specimen was taken from the bronchus' mucosa. The epithelium of this part of the airway is pseudostratified.

Question #: 10

A newborn boy is brought to the neonatal unit because of a 10-minute history of difficulty breathing and cyanosis. His mother has a history of hypothyroidism diagnosed 6 months ago. A diagnosis of neonatal respiratory distress syndrome is made. This condition is seen in infants

with surfactant deficiency. Which of the following functions is most likely associated with this deficient compound?

- A. Secretion of a surface-active agent
- ✓B. Reduction of alveolar surface tension
- C. Production of mucus
- D. Formation of the blood-air barrier
- E. Phagocytosis

Rationale: Surfactant production begins around 20-22 weeks but is present only in small amounts in premature infants and does not reach adequate levels until the late fetal period. Many hormones play a role in the maturation and secretion of surfactant. Thyroxine is one such hormone. Low levels of thyroxine during pregnancy lead to a surfactant deficiency in this case. Type II alveolar cells (pneumocytes) produce surfactant. One of the functions of surfactant is to reduce alveolar surface tension, so that the alveolar walls do not collapse during respiration. Surfactant also aids in trapping foreign particles.

Question #: 11

A 59-year-old man comes to the physician for a follow up examination. He has a 10-year history of emphysema. Which of the following would most likely be expected of the position of the equal pressure point that develops during forced expiration?

- A. It is not affected by static lung compliance
- B. It moves proximally during a forced expiration
- ✓C. It is located distally from the mouth in a patient with emphysema
- D. It is determined by the pulmonary blood flow
- E. It is solely determined by the strength of the ventilatory accessory muscles

Rationale: The equal pressure point (EPP) is the point in the respiratory system where the pleural pressures = airway pressures typically during a forced expiration in a normal person.. Pleural pressures are typically negative but become positive during forced expiration. In normal people the EPP starts in upper cartilaginous airways and then moved down into the lower airways (towards the alveoli) resulting in increased airway resistance and dynamic airway compression. With emphysema, the alveoli are damaged, and the recoil pressures contributing to the airway pressures are less. Thus, airway pressures are lower in emphysema, and the point at which pleural pressure = airway pressures moves further down into the lungs. Indeed, in emphysema, the EPP and dynamic airway compression may even occur with normal respiration.

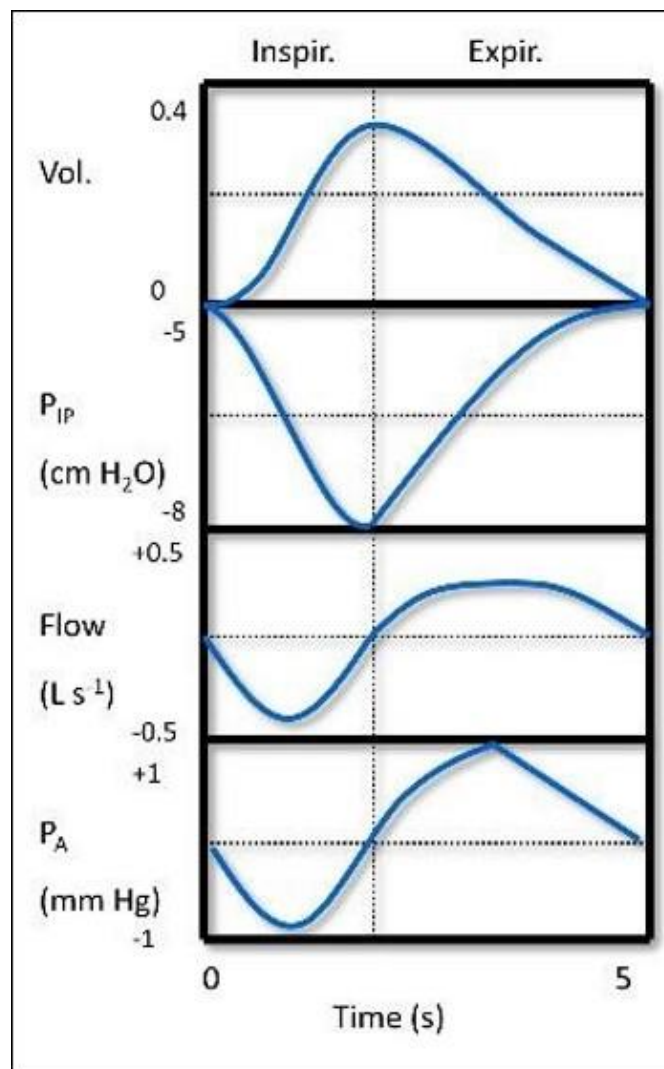
Question #: 12

A 45-year-old woman comes to the physician because of a 2-day history of shortness of breath. She has no significant past medical history. Her temperature is 36.5°C (97.7°F), pulse is 90/min, respirations are 15/min, and blood pressure is 130/83 mm Hg. Her breathing cycle is evaluated, and the results are shown in the attached figure. Which of the following best explains a slight increase in alveolar pressure gradient during expiration compared to that in inspiration?

- A. Contraction of the intercostal muscles
- B. Relaxation of the diaphragm
- ✓C. Decreased radial traction in the airways
- D. Decrease in alveolar volume
- E. Decreased thoracic cavity volume

Rationale: As the PIP becomes less negative, there is less distending force, the radial tension declines and the airway diameter starts to decline; thus, at lower lung volumes, and as we expire, the resistance to airflow is increased.

Attachment:



Breathing cycle_2.jpg

Question #: 13

A 56-year-old man is brought to the emergence department because of a 1-hour history of shortness of breath. He has a 15-year history of myasthenia gravis. His pulse is 88/min and respirations are 22/min with a decreased tidal volume. Physical examination shows weakness of muscles including muscles of breathing. Pulmonary function study shows a decreased change in thoracic volume during breathing cycle and shallow ventilation. Which of the following muscle weakness best explains shallow ventilation in this patient?

- A. Internal intercostal
- ✓B. Diaphragm
- C. External intercostal
- D. Abdominal muscles
- E. Scalene and sternomastoid

Rationale: The diaphragm is the major muscle of inspiration. Contraction of diaphragm increases vertical dimension of the thorax and contributes maximum for ventilation. The increase in volume of the thorax leads to a drop in alveolar pressure, causing air to flow. In this patient, the weakness of diaphragm reduces the change in volume and therefore reduces ventilation (hypoventilation).

Question #: 14

A 28-year-old healthy woman volunteers for a pulmonary study. She is asked to breath normally through a spirometer and pressure measurement shows:

Alveolar pressure (PA) at the beginning of inspiration = 0 cm H₂O,

Intrapleural pressure (PIP) at the beginning of inspiration = -4 cm H₂O,

PA at the end of inspiration = 0 cm H₂O,

PIP at the end of inspiration = -12 cm H₂O,

Which of the following is the most likely transpulmonary pressure at the beginning of inspiration?

- A. -4 cm H₂O
- ✓B. +4 cm H₂O
- C. -8 cm H₂O
- D. +8 cm H₂O
- E. -12 cm H₂O
- F. +12 cm H₂O

Rationale: Transpulmonary pressure is difference between alveolar pressure and intrapleural pressure at a given point in the breathing.

TP Pressure = PA- PIP.

In this case the difference is asked at the beginning of inspiration. Therefore, you must look at the values of PA and PIP at the beginning of inspiration. They are 0 and -4 cm H₂O, respectively. Therefore,

TP = 0 - (-4) = +4 cm H₂O.

Question #: 15

A 34-year-old man is brought to the emergency department 30-minutes after he was involved in a vehicular collision. He sustained multiple rib injuries on the right side of his chest. His pulse is 104/min, blood pressure is 132/84 mm Hg, and respirations are 26/min with shallow tidal volume. Physical examination shows decreased respiratory movements, hyperresonant percussion notes and absent breath sounds in the right upper chest. An x-ray of the chest shows collapse of the right upper lobe of lung. He is diagnosed with right-sided pneumothorax. Measurements of pulmonary pressures on the right side shows an intrapleural pressure of 0 (zero) cm H₂O at the end of expiration. Which of the following is the most likely cause of collapse of right upper lobe of the lung in this patient?

- A. A positive transpulmonary pressure of 10 cm H₂O
- B. Negative intrapleural pressure at the beginning of inspiration
- C. A negative transpulmonary pressure
- ✓D. Zero transpulmonary pressure
- E. Increased recoil of the chest wall.

Rationale: This patient has pneumothorax on the right side. This has produced an intrapleural pressure of 0 cm H₂O at the end of expiration. If the intrapleural pressure is zero, the transpulmonary pressure would also become zero (the difference between alveolar and intrapleural pressures). A positive transpulmonary pressure keeps the lungs inflated in a normal person. When this pressure becomes zero, the lungs collapse, as the elasticity of the lungs makes the lung collapse.

Question #: 16

A 53-year-old woman comes to the physician because of an 8-month history of progressive shortness of breath and fatigue. She does not smoke cigarettes or drink alcoholic beverages. Two years ago, she was diagnosed with lymphoma and had chemotherapy. Her respirations are 22/min with low tidal volume; other vital signs are within normal limits. Physical examination shows no abnormalities. A CT scan of the chest shows fibrosis of the lungs bilaterally. The pulmonary function studies show a flat lung compliance curve. Results of spirometry are shown: Total lung capacity =3.9 Liter, Functional residual capacity =2.2 Liter, Intrapleural pressure at total lung capacity =-25 cm H₂O, Intrapleural pressure at functional residual capacity =-5 cm H₂O. Which of the following values is most likely close to the lung compliance in this patient?

- A. 85 L/cm H₂O
- B. 8.5 L/cm H₂O
- ✓C. 0.085 L/cm H₂O

- D. 0.25 L/cm H₂O
- E. 0.17 L/cm H₂O

Rationale: Lung compliance = change in volume/change in pressure.

In this example,

change in volume = 3.9 - 2.2 = 1.7 Liters.

change in intrapleural pressure = 25-5 = 20 cm H₂O.

Compliance = 1.7/20 = 0.085 L/cm H₂O.

Question #: 17

A 56-year-old man comes to the physician because of a 6-month history of a dry cough and progressive shortness of breath. His respirations are 20/min and shallow; rest of the vital signs are within normal limits. Physical examination shows no abnormalities. Pulmonary function studies show a decreased FEV₁, decreased FVC but an increased FEV₁/FVC ratio. Which of the following mechanisms best explains these pulmonary study findings?

- ✓A. Increased elastic recoil force
- B. Decreased intrapleural pressure
- C. Increased airway resistance
- D. Increased compliance
- E. Decreased chest wall recoil

Rationale: A decrease in FEV₁ and an increase in FEV₁/FVC ratio are characteristic of restrictive lung disease. These concurrent changes can only be explained by increased elastic recoil forces.

Question #: 18

A 59-year-old man comes to the physician because of a 4-week history of coughing and dyspnea. He had similar episodes three times in the last year. He has been smoking 3 packs of cigarettes a day for the past 37 years. His blood pressure is 140/85, pulse is 90/minute, and respirations are 24/minute. Physical examination shows use of accessory muscles for respiration with wheezing and crackles of the lung bases. An x-ray of the chest shows bilateral hyperinflated lungs and a flattened diaphragm. Which of the following is the most likely cause of the patient's presentation?

- A. Asymptomatic asthma
- ✓B. Chronic Bronchitis
- C. Pneumonitis
- D. Interstitial lung disease
- E. Pulmonary Fibrosis

Rationale: The patient most likely has an obstructive disease based on the imaging findings and clinical presentation. The history of smoking is commonly associated with obstructive

diseases of the lung. The obstructive diseases listed are asymptomatic asthma and chronic bronchitis. Due to the presence of symptoms, the patient's age and history of smoking, the best answer is chronic bronchitis.

Question #: 19

A 68-year-old woman comes to the physician because of extreme dyspnea and a productive cough. She has a 60-pack year history of smoking. Physical examinations shows peripheral and central cyanosis and pursed lip breathing. She has been referred to the clinic for measurement of her FRC. Which of the following laboratory studies is most appropriate to measure her FRC?

- A. Bohr's method
 - B. Fowler single breath test
 - C. Flow-Volume spirometry
 - D. Helium dilution test
 - ✓E. Plethysmography
 - F. Volume-Time spirometry
-

Question #: 20

A 13-month-old boy is brought to the physician for follow-up examination because of failure to thrive and anemia. Genetic testing shows that one of the genes encoding beta-globin contains a pathogenic splice site mutation causing loss of the second exon. The second beta globin gene contains a promoter mutation which abolishes expression. Which of the following genetic principles is most likely highlighted in this patient?

- A. Loss of heterozygosity
- B. Locus heterogeneity
- C. Genetic linkage
- ✓D. Allelic heterogeneity
- E. Anticipation
- F. Haploinsufficiency

Rationale: This question is suggesting that the patient has an autosomal recessive blood disorder, and each allele has a different pathogenic mutation, but yet in the same gene. Recall that this is describing compound heterozygote. Loss of heterozygosity typically describes the events which lead to cancer, or uniparental disomy of one chromosome (two identical chromosomes). Locus heterogeneity describes the same disease that can arise from mutation in different, completely independent genes. Genetic linkage describes the association of a known marker with a candidate gene. Anticipation is seen with triplet expansion disorders where an expansion can be seen in children of persons who have the mutation/pre-mutation-expansion. Haploinsufficiency describes the loss of an allele in an autosomal recessive disorder.

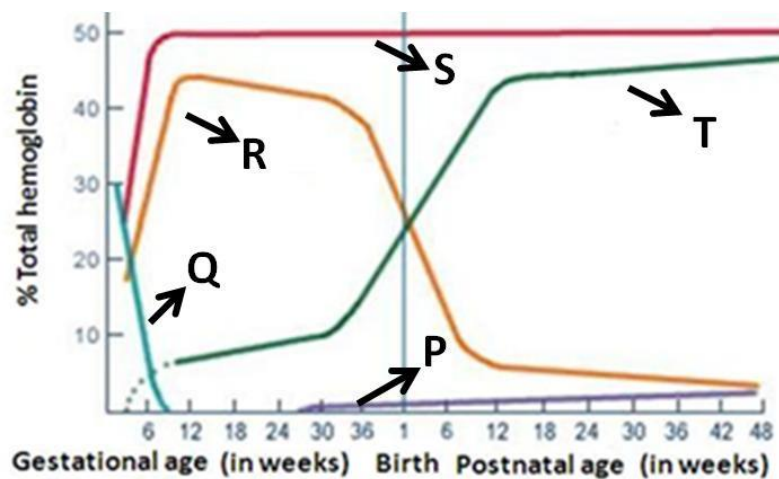
Question #: 21

A 3-year-old girl is brought to the physician by her parents because of a 2-day history of weakness and bone pain following a recent respiratory infection. Physical examination shows conjunctival pallor. Complete blood count shows anemia and hemoglobin electrophoresis shows an abnormal hemoglobin that has a lower mobility towards the anode than HbA. She is prescribed hydroxyurea. The attached figure shows the typical class switching of the globin gene loci. Which of the following lines in the figure represents the gene that is most likely to experience increased expression following treatment with this drug?

- A. The line labeled P
- B. The line labeled Q
- ✓C. The line labeled R
- D. The line labeled S
- E. The line labeled T

Rationale: The attached figure describes the class-switching expression of different isoforms of hemoglobin genes from conception to a few months after birth. Line R represents gamma globin which is used to form Fetal hemoglobin or HbF ($\alpha_2\gamma_2$) that typically declines in expression soon after birth. Hydroxyurea reactivates the expression of gamma globin, and HbF tetramers are formed. It is an FDA approved therapy that has efficacy in preventing acute vaso-occlusive events and can prevent organ damage.

Attachment:



22612 exam soft ID.JPG

Question #: 22

A 1-year-old boy is brought to the physician by his parents because of a 4-week history of repeated bruising. He has a 6-month history of painful joint swellings. Family history of the patient (IV-6) is shown. Laboratory studies show prolonged activated partial thromboplastin time (APTT). Prothrombin time (PT), platelet aggregation tests are within reference range. Which of

the following disorders is most likely in this patient?

- A. Beta-thalassemia
- B. Cystic fibrosis
- C. Familial hypercholesterolemia
- ✓D. Hemophilia A
- E. Sickle cell anemia
- F. Alpha thalassemia

Rationale: The pedigree shows typical X-linked recessive pattern of inheritance. Note only males are affected. Affected males do not have affected children. All daughters of affected males will be carriers of the mutation. Moms of affected sons are carriers of the disorder. Note the skipping of generations - affected children do not have affected parents.

Beta-thalassemia: Autosomal recessive - Characterized by microcytic anemia; HbA levels are low; And HbA2 and HbF levels are elevated.

Cystic fibrosis - Autosomal recessive - Characterized by repeated respiratory infections, and malabsorption and steatorrhea.

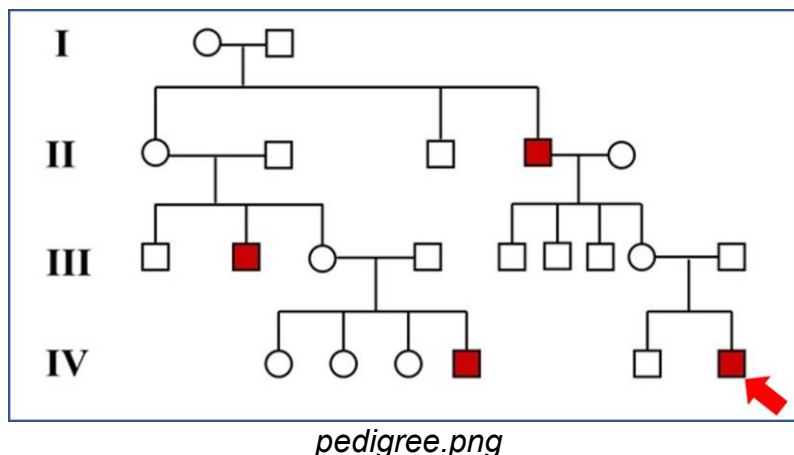
Familial hypercholesterolemia - Autosomal dominant; Characterized by increased risk of atherosclerosis

Hemophilia A - X- linked recessive; A defect in the intrinsic coagulation cascade (Prolonged APTT); The extrinsic coagulation cascade (PT) is normal and platelet aggregation tests are normal

Sickle cell anemia - Autosomal recessive; Characterized by anemia and painful sickling episodes

Alpha thalassemia - Autosomal recessive; Characterized by anemia and low levels of HbA, HbA2 and HbF.

Attachment:



Question #: 23

A 30-year-old woman comes to the physician for a routine examination. She is in her third trimester. Ultrasound of the fetus shows herniation of part of the gut tube into the left hemithorax. A diagnosis of congenital diaphragmatic hernia is made. Which of the following is most likely caused by this condition?

- A. Respiratory distress syndrome
- B. Hemothorax
- C. Pneumothorax
- ✓D. Pulmonary hypoplasia
- E. Eventration of the diaphragm

Rationale: Failure of development of the pleuripitoneal membranes leads to congenital diaphragmatic hernia. The abdominal contents herniates into the thoracic cavity, which takes up the space within this cavity and compresses the contents of this cavity. This results in there being limited space for development of the lungs which leads to pulmonary hypoplasia.

Question #: 24

An autopsy on a 68-year-old man is performed, suspected of smoking is past. The lungs are removed, and large bronchopulmonary lymph nodes are observed. These nodes are most likely closely associated with which of the following structures?

- A. Carina
- ✓B. Hilum of the lung
- C. Main bronchus
- D. Lobar bronchi
- E. Trachea

Rationale: Lymphatic fluid flows from the intrapulmonary nodes to the bronchopulmonary nodes (at the hilum) to the tracheobronchial nodes (at the carina) to the paratracheal nodes to the bronchomediastinal trunk to the right lymphatic duct/thoracic duct back into the venous system.

Question #: 25

A 59-year-old woman comes to the physician because of a 1-month history of a sharp stabbing chest pain during exertion. Physical examination shows pleural rub during auscultation of the chest. A diagnosis of inflammation of the costal pleura is made. Which of the following nerves most likely carries the pain sensation from the affected structure to the CNS?

- ✓A. Intercostals
- B. Phrenic
- C. Vagus
- D. Thoracic splanchnic
- E. Cardiopulmonary

Rationale: The parietal pleura is divided into several regions. The costal pleura is innervated by intercostal nerves. The mediastinal pleura is innervated by the phrenic nerves; The peripheral part of the diaphragmatic pleura is innervated by the intercostal nerves while the central part is

innervated by the phrenic nerves. The pain carried by the phrenic nerve can be referred to the shoulder.

Question #: 26

A 20-year-old man is brought to the emergency department after a motor vehicle collision. His blood pressure is 65/45 mm Hg, pulse is 110/min and respiration are 24/min. Physical examination shows decreased breath sounds over the right lung fields. A diagnosis of tension pneumothorax is made. An emergency decompression is performed, and a chest tube is placed. In which of the following locations would the chest tube be most likely placed?

- A. At the top of the intercostal space
- B. At the middle of the intercostal space
- ✓C. Above the superior border of the rib
- D. Below the inferior border of the rib

Rationale: The chest tube is placed above the superior border of the inferior rib of the intercostal space. This is done to avoid the neurovascular bundle that lies at the superior part of the intercostal space, inferior to the superior rib.

Question #: 27

A 1-day-old newborn boy is brought to the emergency department by his mother because of 1 hour history of not being able to breath. Physical examination shows rapid and labored breathing. Diagnosis of Infant Respiratory Distress Syndrome (IRDS) was made. What is most likely the cause of Respiratory Distress Syndrome?

- A. Pulmonary hypoplasia
- B. Laryngeal atresia
- ✓C. Surfactant deficiency
- D. Polyhydramnios
- E. Tracheal atresia

Rationale: Infant respiratory distress syndrome is the result of a deficiency of surfactant. It is commonly seen in premature infants. Surfactant reduces the surface tension of the alveoli, and if this is deficient, it can lead to the collapse of the alveoli.

Question #: 28

An 18-month-old girl is brought to the physician by her parents because of a 6-hour history of inconsolable crying. She has a history of mild jaundice and intermittent episodes of pallor since

infancy. Physical examination shows conjunctival pallor and mild scleral icterus. Laboratory studies show:

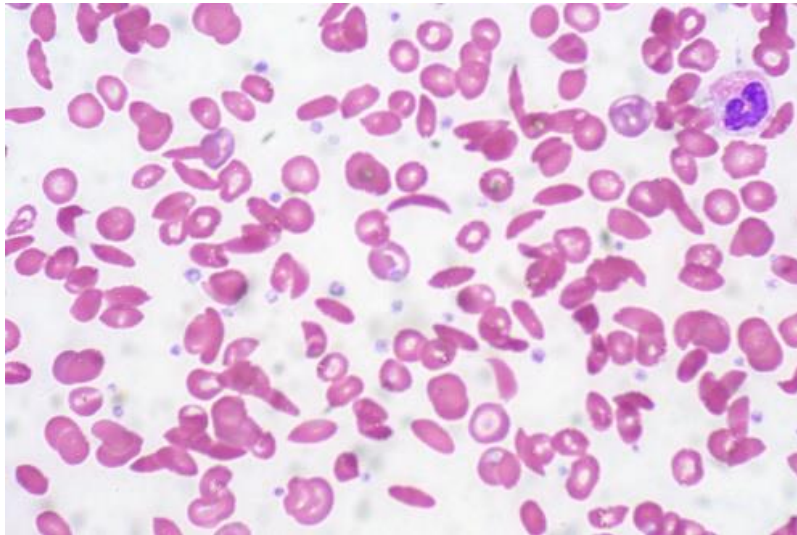
- Hemoglobin: 6.5 g/dL (low)
- Reticulocyte count: 12% (elevated)
- Total bilirubin: 3.2 mg/dL (elevated)

A peripheral blood smear is shown. Which of the following best describes the hemoglobin variant responsible for this patient's presentation?

- A. Higher affinity for oxygen than HbA
- ✓B. Fewer number of negative charges than HbA
- C. Binds 2,3 BPG with lower affinity than HbA
- D. Contains iron in the ferric form
- E. Same primary structure as HbA
- F. Higher electrophoretic mobility towards the anode than HbA

Rationale: The vignette describes a child with painful sickle cell hemolytic crisis. Note the crescent (sickle) shaped RBCs in the peripheral smear. HbS has glutamate (negatively charged) replaced by valine (hydrophobic and neutral) at the 6th amino acid of the beta chain of hemoglobin. As a result, HbS moves slower to the anode than HbA during electrophoresis. HbS and HbA have a similar oxygen affinity and 2,3-BPG binding characteristics. Methemoglobin has iron in the ferric state.

Attachment:



92673.jpg

Question #: 29

A 28-year-old man comes to the physician because of a 2-week history of fatigue, mild dizziness, and shortness of breath on exertion. He recently moved to a mountain village at an elevation of 3,500 meters. He says that he has been feeling progressively more tired over the

past 2 weeks. His temperature is 37°C, pulse is 105/min, and blood pressure is 110/70 mmHg. Laboratory studies show mild hemoconcentration. His hemoglobin shows a sigmoidal oxygen dissociation curve with an increased P50. Which of the following best explains this alteration in hemoglobin function?

- A. Presence of fetal hemoglobin
- B. Increased pH
- C. Low PCO₂
- ✓D. High altitude acclimatization
- E. Hexokinase deficiency
- F. Decreased 2,3-Bisphosphoglycerate
- G. Decreased erythrocyte NADPH levels
- H. Increased erythrocyte glutathione levels

Rationale: The ODC is shifted to the right with an increase in P50 when the hemoglobin oxygen affinity is reduced. This occurs when the pH is low (acidosis), high levels of 2,3-BPG (high altitude acclimatization and pyruvate kinase deficiency) and high PCO₂.

The Hemoglobin ODC is shifted to the left with a fall in P50 when the oxygen affinity of hemoglobin is increased. This occurs in the presence of fetal Hb, low levels of 2,3-BPG (hexokinase deficiency).

Note that NADPH and glutathione levels have no effect on the ODC.

Question #: 30

A 12-year-old boy is brought to the physician by his parents because of a 6-month history of fatigue and generalized weakness. Physical examination shows a pale complexion and conjunctival pallor. Laboratory studies show:

- Hemoglobin: 8.0 g/dL (low)
- Hematocrit: 28% (low)

The oxygen dissociation curve plotted from his hemoglobin is shown. Which of the following best explains his altered hemoglobin-oxygen binding affinity?

- A. Pyruvate kinase deficiency
- B. High altitude adaptation
- ✓C. Hexokinase deficiency
- D. Carbon monoxide poisoning
- E. Sickle cell disease

Rationale: The Hb ODC is shifted to the left (become more hyperbolic). This is observed in hexokinase deficiency. In patients with hexokinase deficiency, there is decreased formation of 2,3-BPG in RBCs. As a result, hemoglobin has a higher affinity for oxygen.

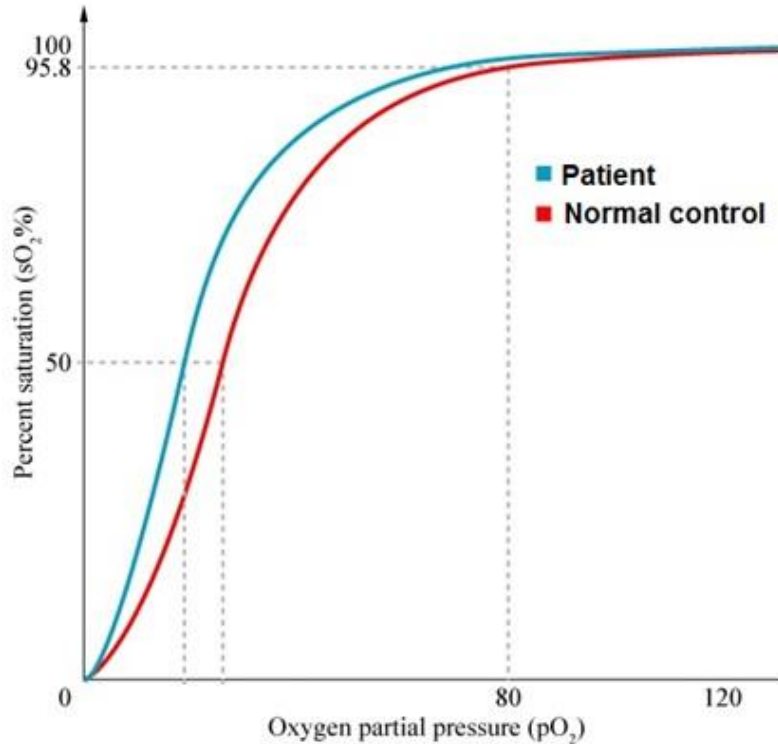
In pyruvate kinase deficiency, the Hb ODC is shifted to the right. There is increased formation of 2,3-BPG in pyruvate kinase deficiency. 2,3-BPG binds to Hb and reduces Hb oxygen affinity and

increases unloading of oxygen.

In high altitude adaptation, there is a shift of Hb ODC to the right which decreases oxygen affinity of Hb. Hypoxia stimulates increased production of 2,3-BPG.

In carbon monoxide poisoning, the number of binding sites for oxygen is reduced and the HbODC becomes hyperbolic. CO stabilizes the R state of Hb and prevents oxygen unloading.

Attachment:



170500.jpg